

Neuroscientific Frontiers — Module 02: Agents —

Study Questions

1. What is the default mode network (DMN)? What does it do during rest? How does Active Inference reframe "rest" as active self-modeling?
2. Which DMN regions correspond to which aspects of the self-model? Map mPFC, PCC, TPJ, hippocampus, and precuneus to specific self-model functions.
3. How does interoceptive inference ground subjective experience? What is the interoceptive hierarchy from periphery to anterior insula?
4. Explain how emotions are interoceptive prediction errors. What does anxiety "feel like" in terms of interoceptive predictions? How does this differ from the James-Lange theory?
5. What is the rubber hand illusion? How does it demonstrate that body ownership is an inference rather than a fixed property?
6. How do phantom limb sensations support the predictive model of body ownership? Why does the prophylactic model persist after amputation?
7. What neural evidence connects the anterior insula to the "sentient self"? How do heartbeat evoked potentials (HEPs) relate to interoceptive accuracy?
8. Compare depersonalization, Cotard's syndrome, and alien hand syndrome as distinct self-model failures. What specific component of the self-model fails in each?
9. How does anosognosia (denial of deficit after stroke) illustrate the failure of self-model updating? Why do priors about one's own abilities resist prediction error?
10. What is the "minimal self" (Zahavi)? How does it correspond to the most basic level of interoceptive inference?
11. How does anesthesia provide evidence for the self-model theory of consciousness? What is disrupted during loss of consciousness — sensory processing or self-modeling?
12. How does the dreaming state challenge and support the self-model theory? The self persists in dreams but is decoupled from reality — what does this reveal?
13. How does the self-model develop across the human lifespan? Map neonatal, infant, childhood, and adolescent self-model development onto neural maturation.
14. How might meditation systematically alter the self-model? What neuroscientific evidence exists for meditation-induced changes in DMN and insular activity?

15. How does alexithymia (difficulty identifying emotions) relate to interoceptive precision? What specific neural markers differentiate alexithymic from non-alexithymic individuals?
16. Connect the proprioceptive body schema to action planning: how does knowing where your body IS enable predicting where it WILL BE?
17. How does the Active Inference self-model account compare to Damasio's somatic marker hypothesis? Where do they converge and diverge?
18. What would a self-model without narrative (without autobiographical memory) look like? Does amnesia eliminate the narrative self while preserving the minimal self?
19. How might psychedelics alter the self-model? What evidence connects 5-HT2A agonism to self-model dissolution ("ego death")?
20. Critically evaluate: Is the "self as inference" framework falsifiable? What experiment could definitively test whether the self is a prediction rather than a percept?